



Max. Smart Timer

Instituto Salesiano Macau

F5B

Jason Chong 06

Kevin Chan 01

Jerry Vong 30

Content

1. Project Summary	3
1.1 Technical Overview (for technical readers)	4
1.2 Technical Overview (For non-technical readers)	5
2. Design Background, Motivation, and Objectives	6
2.1 Background and Motivation	6
2.2 Data Collection and Research	6
2.3 Comparison with Existing Solutions	7
2.4 Project Objectives and Success Criteria	8
2.5 Creativity and Social Value	8
3. Technical explanations:	9
3.1 Design	9
3.1.1 Welcome system	9
3.1.2 Gear system	9
3.2 Dimensions	12
4. Results	13
4.1 Comparison to the original plan	13
4.2 Our process	13
4.3 Our product	14
5. Conclusion	16
5.1 Goal check	16
5.2 Future Development	16
6. References	17

1. Project Summary

Max. Smart Timer is an innovative desktop device inspired by the cockpit dashboard of a Formula 1 racing car, designed to help university students and late-night workers maintain focus, manage time, and create a more engaging study environment.

The device integrates several practical and entertaining functions into a single compact unit: a precise digital clock synchronized via the internet, real-time indoor temperature and humidity display, customizable ambient RGB lighting that surrounds the entire structure, and a unique interactive feature—an automatically triggered mini RC racing car that begins circling a small track when the user inserts a symbolic “key” into a holder. This action visually represents the start of a focused work or study session (e.g., Pomodoro mode), adding a motivational and playful element to routine productivity tasks.

The core hardware components include:

- XIAO ESP32 C3 microcontroller (main processing and Wi-Fi connectivity)
- 4-inch TFT LCD monitor
- Arduino UNO
- 2.42-inch SSD1309 OLED display
- DHT11 temperature/humidity sensor
- WS2812B addressable RGB LED strip
- IR sensor
- Push Button
- Modified Mini F1 RC Drift Car with new DC motor control circuit

The overall appearance mimics an F1 instrument panel, making it both a functional timer and an eye-catching desk accessory that helps reduce stress, combat eye fatigue from harsh lighting, and improve time management through gamified interaction.

1.1 Technical Overview (for technical readers)

The system leverages the ESP32's built-in Wi-Fi module to connect to NTP servers for accurate real-time clock synchronization and supports timer modes such as Pomodoro. The 0.96" OLED screen (128×64 resolution, I²C interface) is driven using the Adafruit GFX and SSD1306 libraries to display time, date, temperature, humidity, and status messages.

Environmental data is acquired from the DHT11 sensor and rendered live on the display. Ambient lighting is achieved with a WS2812B LED strip controlled via the FastLED library, allowing dynamic color, brightness, and pattern customization (currently button-controlled, with potential for future app integration).

A key detection mechanism uses a micro switch connected to an ESP32 GPIO pin. Upon key insertion, the microcontroller activates a newly designed RC car control circuit using an L298N dual H-bridge motor driver module. PWM signals from the ESP32 drive the car's dual DC motors, enabling automatic forward motion and simple looping behavior around a small 3D-printed track for at least 30 seconds.

The entire assembly is enclosed in a custom 3D-printed F1-style housing that integrates the key holder, display, LEDs, and car track. The firmware is developed in the Arduino IDE, prioritizing low power consumption, efficient multitasking, and reliable wireless operation—making this project significantly more interactive and feature-rich than typical ESP32-based clock implementations.

1.2 Technical Overview (For non-technical readers)

The Max. Smart Timer is an innovative desktop device inspired by the dashboard of a Formula 1 racing car, designed to help university students and late-night workers manage time, maintain focus, and liven up their study environment. Many people struggle with monotonous routines, distractions, and eye strain during long working hours. This smart timer tackles those common pain points by combining practical daily tools with a playful, interactive experience.

On the practical side, the device features a highly accurate digital clock, a real-time indoor temperature and humidity display, and soft, customizable ambient lighting to help reduce eye fatigue. However, its standout feature is a unique "race start" ritual. When you are ready to begin a focused work session, you insert a physical key into the device's holder. This action instantly triggers a miniature racing car to speed around a built-in track. By turning the simple act of starting a timer into a fun, gamified event, the Max. Smart Timer transforms a stressful chore into a highly motivating and rewarding experience.

2. Design Background, Motivation, and Objectives

2.1 Background and Motivation

Many university students, particularly in high-pressure academic environments such as Macau, face common challenges during late-night study sessions: difficulty maintaining consistent focus, monotonous surroundings, eye strain from poor lighting, and ineffective time management. Traditional timers or phone alarms often lack the emotional or visual engagement needed to sustain motivation over extended periods.

This project was conceived to address these pain points by combining practical utility (accurate timekeeping, environmental monitoring, soft ambient lighting) with an entertaining, gamified interaction inspired by the excitement of Formula 1 racing—turning the act of starting a study session into a small but satisfying “race start” ritual.

2.2 Data Collection and Research

The design draws heavily on open-source hardware projects published on Instructables, including the original “F1 Cockpit Inspired Digital OLED Clock with ESP32” (for the base clock/display inspiration), “Mini F1 RC Drift Car” (for RC chassis and motor concepts), and other related 3D-printable and sensor-based builds.

Official datasheets for the Arduino UNO, DHT11 sensor, WS2812B LEDs, 2.42-inch SSD1309 OLED, L298N motor driver, and Seeed studio XIAO ESP32-C3 module were consulted to confirm electrical compatibility, power requirements, and communication protocols (I²C, single-wire, PWM). Component availability, pricing, and performance were cross-checked via Taobao listings to ensure feasibility within a student budget.

Preliminary breadboard prototypes verified that all modules operate reliably on a shared 5V supply with sufficient GPIO pins remaining on the ESP32.

2.3 Comparison with Existing Solutions

The original F1-inspired OLED clock project provides only a time display with no interactive or environmental features. Commercial smart clocks (e.g., those with weather and alarms) are significantly more expensive, lack customization, and do not include physical motivational elements.



Fig 2.3 Clock found on the Internet

Other maker projects, such as smart reading easels or motorized winches, focus on single-purpose utility without the multi-sensory, gamified experience offered here. By integrating weather monitoring, adjustable RGB mood lighting, and the unique key-triggered RC car start, Max. Smart Timer delivers higher engagement and lifestyle relevance at a fraction of commercial cost (<200 CNY estimated total).

2.4 Project Objectives and Success Criteria

The project is considered successful if the following core functionalities are reliably achieved:

1. Accurate NTP-synchronized clock and configurable timer displayed on the OLED screen
2. Real-time temperature and humidity readings from DHT11, clearly shown alongside time
3. Fully controllable RGB ambient lighting (color, brightness, patterns) surrounding the device
4. Key insertion reliably detected, triggering the RC car to automatically run around the track for at least 30 seconds
5. Complete integration within a cohesive, 3D-printed F1 cockpit-style enclosure with stable wireless connectivity and no major functional bugs

Meeting all five criteria constitutes project success.

2.5 Creativity and Social Value

This project goes beyond simply combining off-the-shelf sensors and microcontrollers. It responds to a real, researched student pain point—lack of motivation and harsh late-night study conditions—by introducing a novel “race-start” interaction metaphor that gamifies the beginning of focused work sessions.

The design demonstrates empathy for campus life challenges and promotes healthier time-management habits among young people through entertainment and visual reward, contributing modest but meaningful social value.

3. Technical explanations:

3.1 Design

3.1.1 Welcome system

As mentioned, our thoughts are to detect when the owner of the house comes back. As he comes back, the placement of the key will be detected and send a signal to the gear system. Subsequently, the F1 car will start to run and welcome the owner.

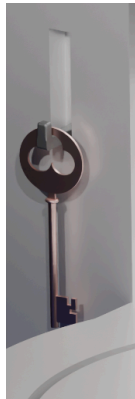


Fig 3.1.1.1 Key detector

3.1.2 Gear system

Our gear system uses a planetary gear for the movement and rotation of the tiny F1 car. The power of this system comes from a 37GB-520 AC motor (144RPM).

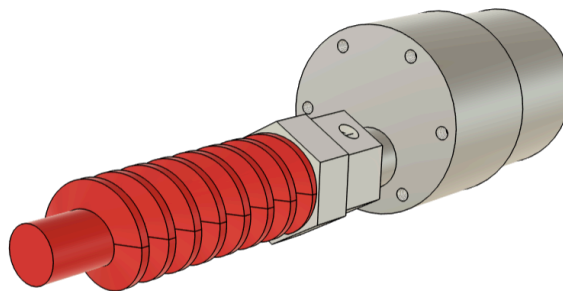


Fig 3.1.2.1 Connected worm screw with motor

The system consists of a planetary gear, helical herringbone sun gear, and worm gears. The worm gear, which is used to transfer energy from the horizontal direction (x-axis) to the vertical direction (z-axis), has 15 teeth with an angle of 10 degrees, while the sun gears have 62 teeth with a helix angle of 30 degrees, and the planetary gear includes 30 gears with the same angle. The worm gear is connected to the sun gear vertically.

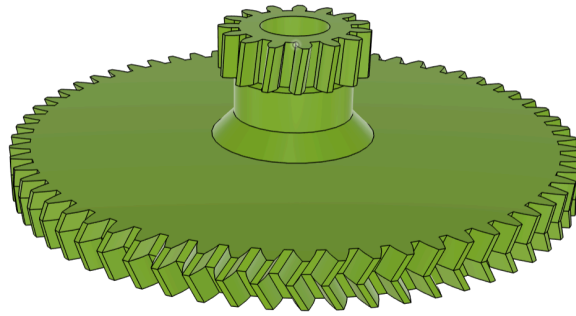


Fig 3.1.2.2 A gear integrated with a worm gear helical herringbone sun gear

All components of the system work in union, resulting in a steady circular motion of the F1 car with a radius of 92mm. Furthermore, to maintain the front of the car's direction, a little guide track is used.

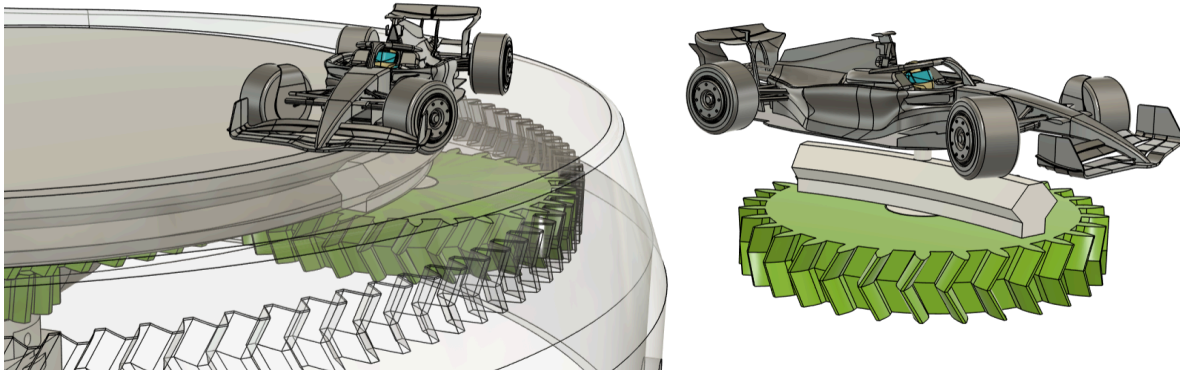


Fig 3.1.2.2 F1 car connected to guide track.

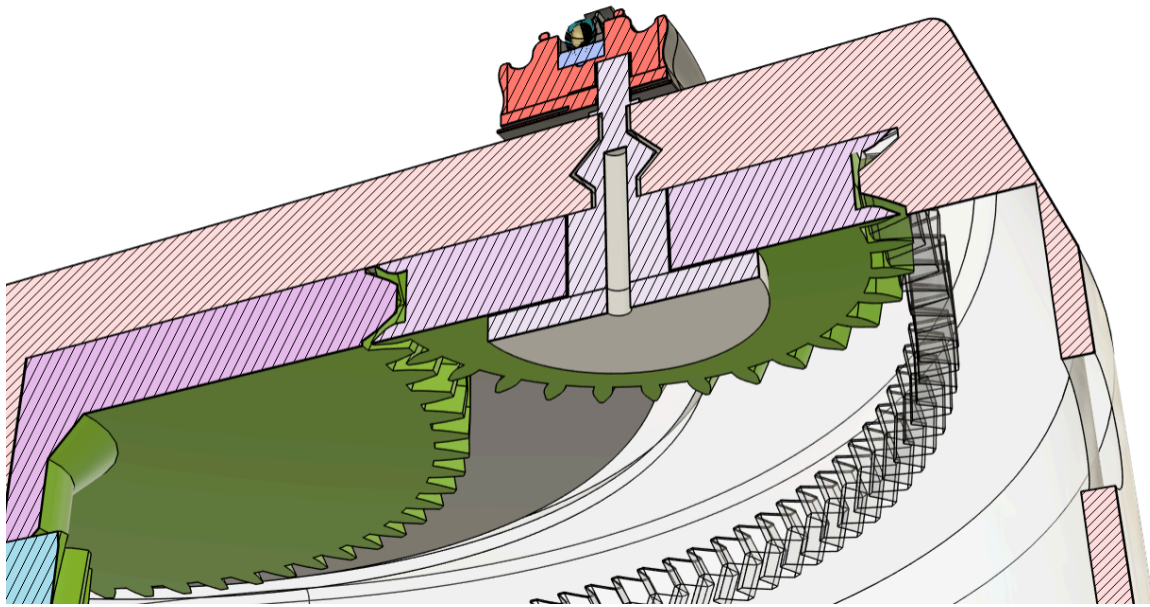


Fig 3.1.2.3 Section analysis of the guide track

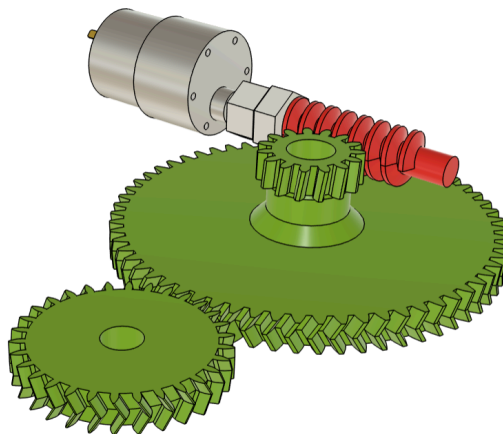


Fig 3.1.2.4 Final gear system

3.2 Dimensions

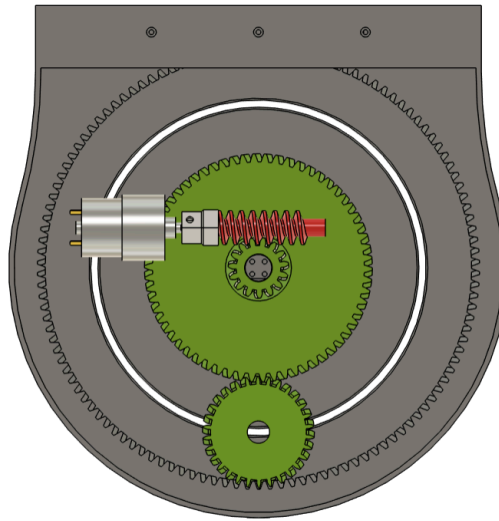


Fig 3.2.1 The bottom view of our product (in CAD)

Base: $\varnothing 28.0 \text{ mm} \times 7.0 \text{ mm}$ tall

Clock: $25.0 \text{ mm} \times 3.7 \text{ mm} \times 13.0 \text{ mm}$

Key Holder: $8.0 \text{ mm} \times 4.0 \text{ mm} \times 20.0 \text{ mm}$

Overall: $344.63 \text{ mm} \times 1155.76 \text{ mm} \times 341.21 \text{ mm}$

4. Results

4.1 Comparison to the original plan

To finish this project is definitely easier said than done; what shows up first is the difference in the dimensions. What we have planned is definitely smaller, with fewer components. The second is the complexity, which took longer than we anticipated, including more coding and sophisticated multi-component adjustment and collaboration

4.2 Our process

To explain our management over time, we started off by constructing the basic coding frame of the entire project, along with working on the 3D model for the product. At the same time, the third member marks down the process and starts working on the documents, including the PowerPoint, proposal, and the PowerPoint.

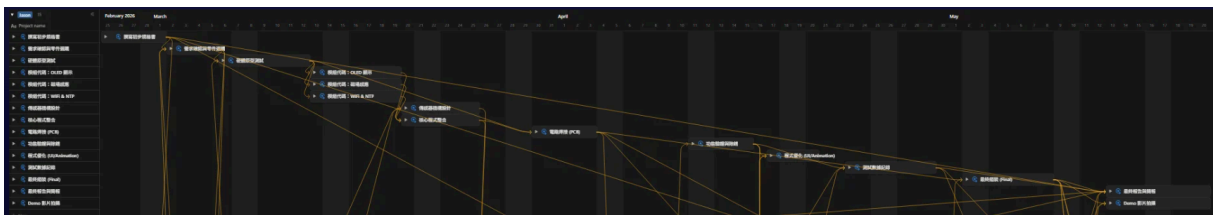


Fig 4.2.1 Jason's Gantt Chart



Fig 4.2.2 Kevin's Gantt Chart

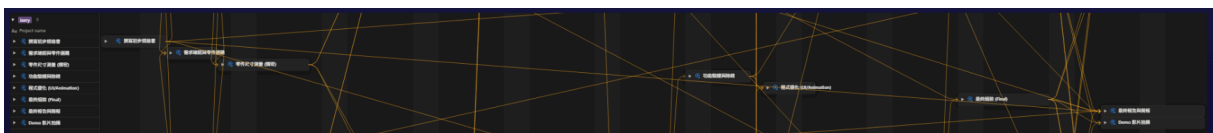


Fig 4.2.3 Jerry's Gantt Chart

Our initial work schedule is planned as above, but at different times, we all have different events stopping us from attending lessons. Despite the disturbance, we managed to reschedule and maintain the overall progress, thus being able to finish the project in time.

Milestones:

- On 11/3/2026, we started great, with a final decision on the exterior
- On 8/4/2026, we finished the systems and began to focus on the hardware.
- On 22/4/2026, we completed the project and shifted our focus to the documents.

4.3 Our product

- This is our final product, on the right is a key sensor, which uses an IR sensor do detect whether a key is present, if so, it activates the gear system, which is put in the front of the product, causing the car to start running, the back is where the clock system is embedded in, displaying various information related to the time and weather, with the shape of the clock inspired from the steering wheel and the dashboard of the F1 racing car.

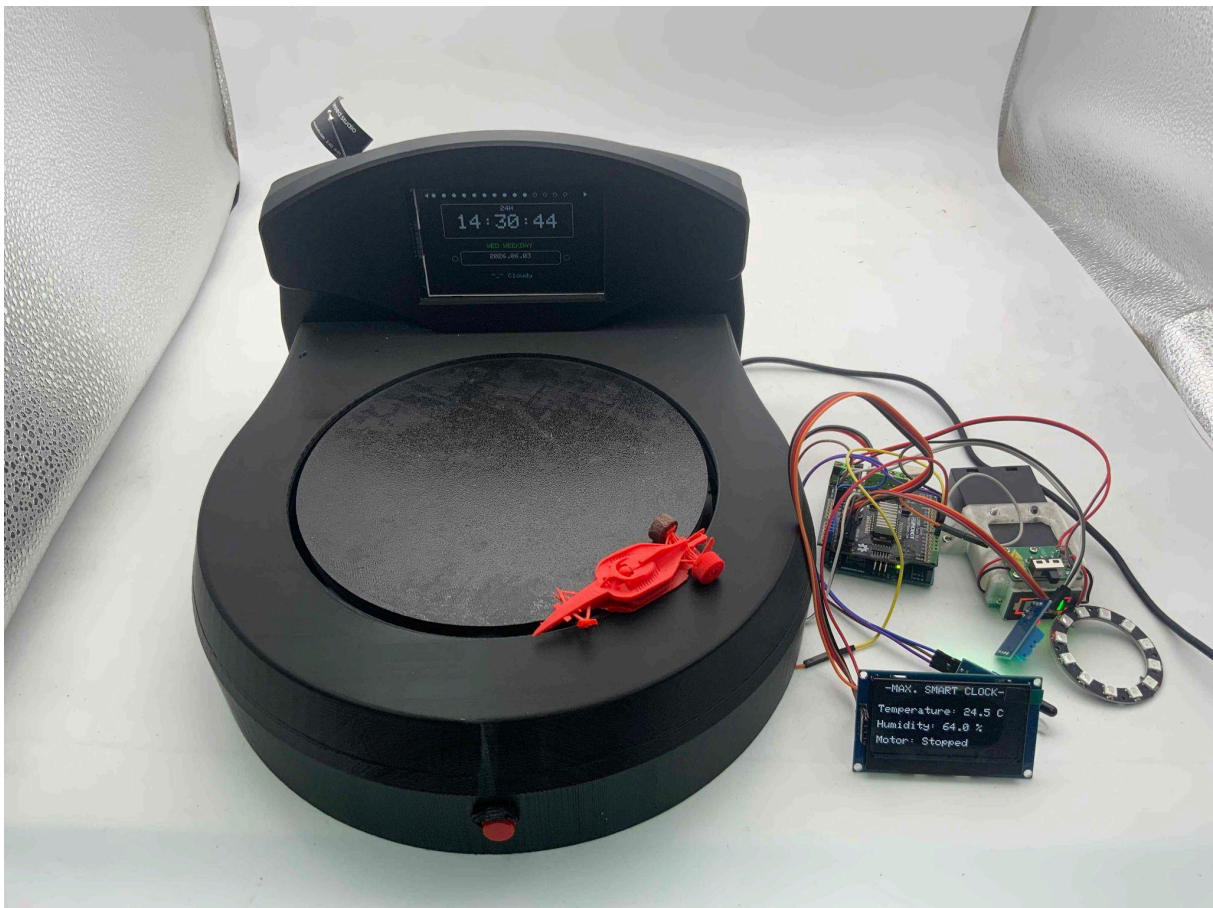


Fig 4.3.1 Final product



Fig 4.3.2 Final product by rendering

5. Conclusion

5.1 Goal check

According to the goals set in 2.4, we can declare that our project has been a success, including an accurate NTP-synchronized clock and configurable timer displayed on the OLED screen, while the DHT 11 sensor can update the temperature and humidity in real-time, and reliably detect the key insertion reliably detected, triggering the RC car to automatically run around the track for at least 30 seconds. Although we failed to have a fully controllable RGB ambient lighting (color, brightness, patterns) surrounding the device, we still managed to make a complete integration within a cohesive, 3D-printed F1 cockpit-style enclosure with stable wireless connectivity and no major functional bugs.

5.2 Future Development

To further develop our project, we have brainstormed multiple possible extensions, including a stopwatch feature and various themes by making changes within the OLED screen and the main circuit board, and a wireless emergency alert system by integrating related API and adding a buzzer.



Fig 5.2 Wireless Emergency Alert System

6. References

- Instructables: “F1 Cockpit Inspired Digital OLED Clock With ESP32” (primary base project)
<https://www.instructables.com/F1-Cockpit-Inspired-Digital-OLED-Clock-With-ESP32/>
- Instructables: “Mini F1 RC Drift Car” (RC car chassis and motor reference)
<https://www.instructables.com/Mini-F1-Rc-Drift-Car/>
- Instructables: “Reading Easel” (comparison project)
<https://www.instructables.com/Reading-Easel-1/>
- Instructables: “3D Printed Electric Winch” (3D printing inspiration)
<https://www.instructables.com/3D-Printed-Electric-Winch-Strong-Enough-to-Lift-Me/>
- Official documentation: ESP32 NTP client examples, Adafruit SSD1306 & GFX libraries, FastLED library (Arduino.cc, GitHub)
- DHT11 sensor datasheet
- Taobao component listings and price comparisons (ESP32-WROOM-32, 0.96" OLED, WS2812B, L298N, etc., current 2026 pricing)